



Team Name
TEKKUZEN

Theme
EDUCATION
AI-Powered RAG Tutor for College Courses

Team Members' Details

Name	Year	College
 Nandini Paul	3rd	St. Thomas' College of Engineering and Technology
Name	Year	College
 Ayush Chowdhury	2nd	St. Thomas' College of Engineering and Technology
Name	Year	College
 Rajdeep Das	2nd	St. Thomas' College of Engineering and Technology
Name	Year	College
 Vikash Kr Gupta	2nd	St. Thomas' College of Engineering and Technology



PROPOSED SOLUTION

- ⚙️ **Explanation:**
Course-aware AI tutor that uses RAG to answer questions only from approved lecture materials.
- ⚙️ **Addressing the Problem:**
Students often rely on AI tools that provide incorrect or irrelevant answers and directly solve homework, harming real learning.
Our solution preserves academic integrity by grounding every response in course content and guiding students step-by-step using Socratic hints instead of direct solutions.
- ⚙️ **Innovation:** Department-Wise Content **Isolation**, **Confidence-Based Refusal System**, **Exam Mode Lock**, Faculty-Controlled AI Scope



TECHNICAL APPROACH

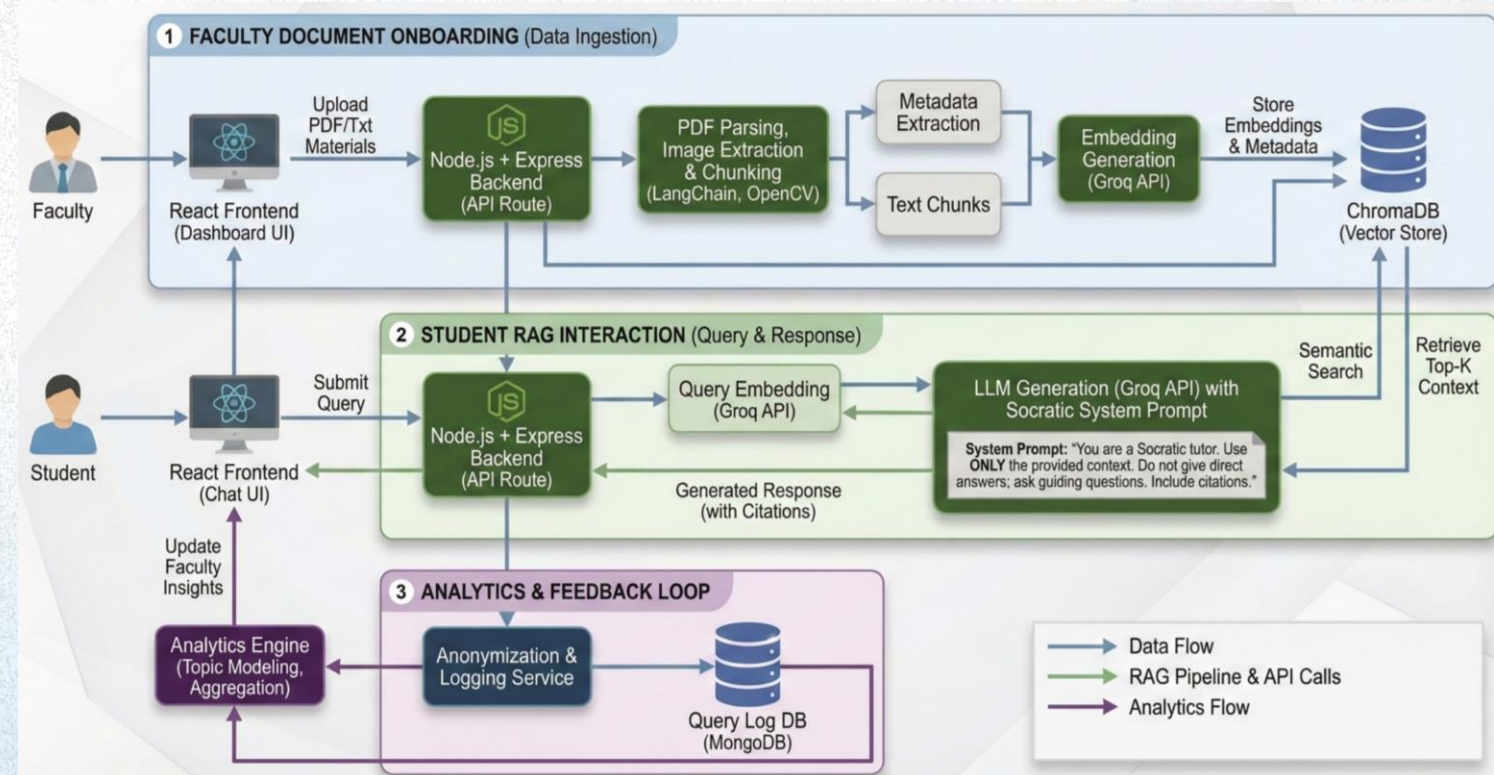
⚙️ The Arsenal:

Node.js + Express, Groq API, ChromaDB Vector DB, RAG Pipeline, PDF Parsing, Semantic Search, Socratic Prompting, React Frontend, Docker Deployment.

⚙️ The Execution:

- Faculty upload → OpenCV extraction
- Chunking → Embeddings → ChromaDB
- Student query → Retrieval → AI response
- Permission enforcement
- Analytics logging

⚙️ Blueprint of Brilliance:





FEASIBILITY & VIABILITY

❄ Implementation Feasibility:

- Use ready-made API — no model training needed
- Node.js backend + simple React frontend
- ChromaDB for fast vector storage
- OpenCV + OCR for page extraction
- Core RAG pipeline can be built within hackathon time

❄ Identifying Challenges:

- Noisy OCR output from scanned PDFs
- Hallucinated answers from AI
- Slow response time with large PDFs
- Students trying to bypass permissions
- Limited GPU / Compute resources

❄ Strategic Solutions:

- Preprocess pages using OpenCV for better OCR accuracy
- Strict RAG-based context filtering
- Source-cited answers only
- Department-wise content isolation
- Cache frequent queries
- Exam-mode hint-only lock



THE RIPPLE EFFECT: TRANSFORMING THE LANDSCAPE



⚙️ **Target Impact:**

- Working prototype of Course-Aware AI Tutor
- Faculty uploads notes → students ask doubts
- Real-time, source-cited AI responses
- Reduced confusion for students
- Faster doubt resolution without teacher overload
- Live demo of faculty dashboard analytics

⚙️ **Broader Influence:**

- Scalable AI tutor for all colleges and departments
- Improved learning quality and academic integrity
- Reduced faculty workload for repetitive doubts
- Affordable AI education tool for rural colleges
- Data-driven teaching improvements
- Multilingual support for Indian students
- Offline campus deployment (privacy-first)
- Long-term cost reduction in tutoring
- Greener learning via paperless notes